

Question: Write a program for linear search.

PROGRAM:

```
1 #include<stdio.h>
2 int main(){
3     int i,j,n,key;
4     int a[20];
5     int flag=0;
6
7     printf("Enter the size of the array:\n");
8     scanf("%d",&n);
9
10
11     printf("Enter the elements of the array:\n");
12     for(i=0;i<n;i++){
13         scanf("%d",&a[i]);
14     }
15
16
17     printf("enter the element to search:\n");
18     scanf("%d",&key);
19
20
21     for(i=0;i<n;i++){
22         if(a[i]==key){
23             printf("Element found at %d position.\n",i);
24             flag=1;
25             break;
26         }
27     }
28     if(flag==0){
29         printf("Element not found.\n");
30     }
31     return 0;
32 }
33
34
35
```

TERMINAL PROCESS:

```
srmap@CV501-CSELAB-SYS16-U:~/Desktop$ cd AP25110010976
srmap@CV501-CSELAB-SYS16-U:~/Desktop/AP25110010976$ gcc linear.c -o li
srmap@CV501-CSELAB-SYS16-U:~/Desktop/AP25110010976$ ./li
Enter the size of the array:
6
Enter the elements of the array:
12
15
18
32
23
65
enter the element to search:
32
```

TERMINAL OUTPUT:

Element found

2nd

Program _____

Question: Write a program for binary search.

PROGRAM:

```
1 #include<stdio.h>
2 int main(){
3     int i,n,key;
4     int a[20];
5     int flag=0;
6
7     printf("Enter the size of the array:\n");
8     scanf("%d",&n);
9
10
11    printf("Enter the elements of the array:\n");
12    for(i=0;i<n;i++){
13        scanf("%d",&a[i]);
14    }
15
16
17    printf("enter the element to search:\n");
18    scanf("%d",&key);
19
20    int beg=0,end=n-1;
21    int mid=(beg+end)/2;
22
23
24    while(beg <= end){
25        mid = (beg + end)/2;
26        if(a[mid] == key){
27            printf("Element found at position %d\n", mid + 1);
28            flag = 1;
29            break;
30        }
31        else if(a[mid] < key){
32            beg = mid + 1;
33        }
34        else{
35            end = mid - 1;
36        }
37    }
38    if(flag == 0) {
39        printf("Element not found\n");
40    }
41    return 0;
42 }
43
```

TERMINAL PROCESS (INPUT AND OUTPUT):

```
srmap@CV501-CSELAB-SYS16-U:~/Desktop/AP25110010976$ gcc bin.c -o bin
srmap@CV501-CSELAB-SYS16-U:~/Desktop/AP25110010976$ ./bin
Enter the size of the array:
5
Enter the elements of the array:
1
34
54
675
876
enter the element to search:
54
Element found at position 3
srmap@CV501-CSELAB-SYS16-U:~/Desktop/AP25110010976$
```